

1(a)(i)	A mitochondrion ; B <u>rough endoplasmic reticulum</u> ; A ribosome / 80S ribosome	2
1(a)(ii)	any three from: (more efficient) synthesis / metabolism / AW, of named product ; e.g. lipid / named lipid phospholipid steroid / steroid hormone / named, e.g. oestrogen / testosterone sterol / cholesterol I ref. to storage other function ; e.g. detoxification max two from: <i>idea that</i> synthesis of products / AW, is a pathway / requires more than one enzyme ; further detail ; e.g. enzymes of same pathway are, close together / in clusters products of one reaction are substrates of the next <i>suggestion that</i> membrane gives enzymes, more stability / greater protection (from degradation) ; AVP ; e.g. <i>idea of</i> greater efficiency as enzyme held so that active site faces substrates in lumen <i>suggestion that</i> allows reactions to occur on cytoplasmic side of membrane <i>suggestion that</i> allows reaction to continue / complete, in SER transport vesicles	3

Question	Answer	Marks
1(b)	any three from: higher magnification (without loss of detail) ; higher resolution / ability to see object that are close together as separate objects ; <i>ref. to resolution of (up to) 0.5 nm compared to, 200 nm / 0.2 μm (light microscope) ;</i> A ranges, e.g. 0.5–30 nm and 100–300 nm ability to see organelles such as, SER / ribosomes / other named organelle ; AVP ; e.g. very thin sections, qualified, e.g. for greater internal detail	3

Question	Answer	Marks
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